

SERIE DE FOURIER

Ing Leandro Borgnino

Teoría de Señales y Sistemas Lineales

Guía Prácticos 3

Resoluciones por Narella Cerquetti.

Tema 1: Serie de Fourier

(1) Determinar la frecuencia fundamental ω_0 y los coeficientes a_k de la Serie de Fourier.

1.

$$x(t) = \text{sen}(3\pi t)$$

$$a_1 = \frac{1}{2j}, \quad a_{-1} = -\frac{1}{2j}$$

2.

$$x(t) = 2 + \cos\left(\frac{2\pi}{3}t\right) + 4\text{sen}\left(\frac{5\pi}{3}t\right)$$

$$a_0 = 2, \quad a_2 = a_{-2} = -\frac{1}{2}$$
$$a_5 = -j2, \quad a_{-5} = j2$$

3.

$$x(t) = \begin{cases} 1.5, & 0 \leq t < 1 \\ -1.5, & 1 \leq t < 2 \end{cases}$$

con frecuencia $\omega_0 = \pi$

$$a_k = -j3 \frac{\text{sen}^2(k\pi/2)}{k\pi}$$

4.

$$x(t) = \cos(4t) + \cos(7t)$$

$$a_4 = a_{-4} = a_7 = a_{-7} = \frac{1}{2}$$

5.

$$x(t) = 1 + \frac{1}{2}\cos(2\pi t) + \cos(4\pi t) + \frac{2}{3}\cos(6\pi t)$$

$$a_0 = 1, \quad a_1 = a_{-1} = \frac{1}{4}$$
$$a_2 = a_{-2} = \frac{1}{2}, \quad a_3 = a_{-3} = \frac{1}{3}$$

6.

$$x(t) = \begin{cases} 1 & : |t| < T_1 \\ 0 & : T_1 < |t| < T/2 \end{cases}$$

$$a_k = \frac{\text{sen}(kT_1 2\pi)}{k\pi}$$

7.

$$x(t) = \cos\left(\frac{2\pi}{3}t\right) + \text{sen}\left(\frac{2\pi}{3}t + \frac{\pi}{8}\right)$$

$$a_1 = \frac{1}{2} + \frac{1}{2j}e^{j\pi/8}$$

$$a_{-1} = \frac{1}{2} - \frac{1}{2j}e^{-j\pi/8}$$

8.

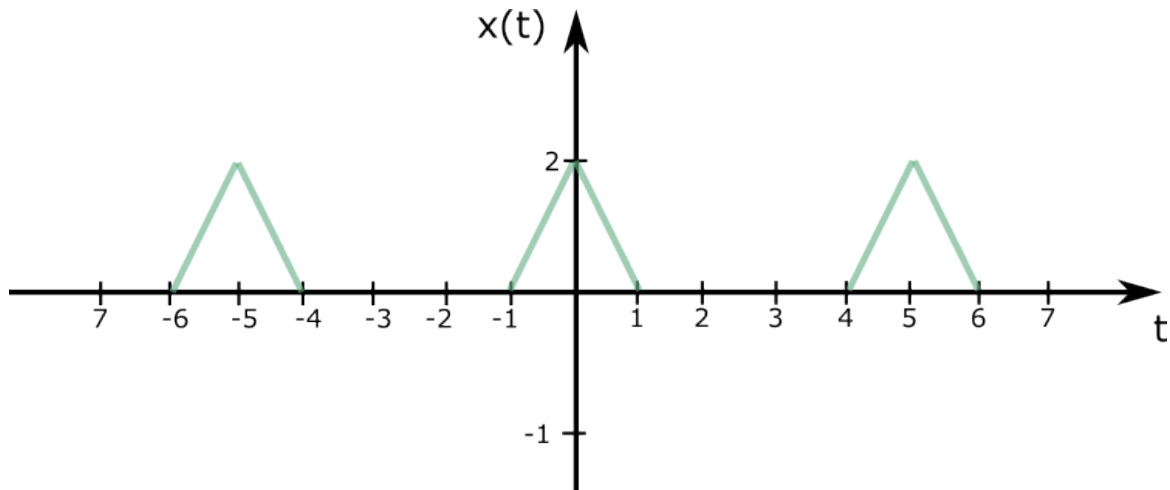
$$x(t) = \cos\left(\frac{\pi}{8}t + \frac{\pi}{6}\right) + \text{sen}\left(\frac{\pi}{4}t - \frac{\pi}{3}\right)$$

$$a_1 = \frac{1}{2}e^{j\pi/6}, \quad a_{-1} = \frac{1}{2}e^{-j\pi/6}$$

$$a_2 = \frac{1}{2j}e^{-j\pi/3}, \quad a_{-2} = -\frac{1}{2j}e^{j\pi/3}$$

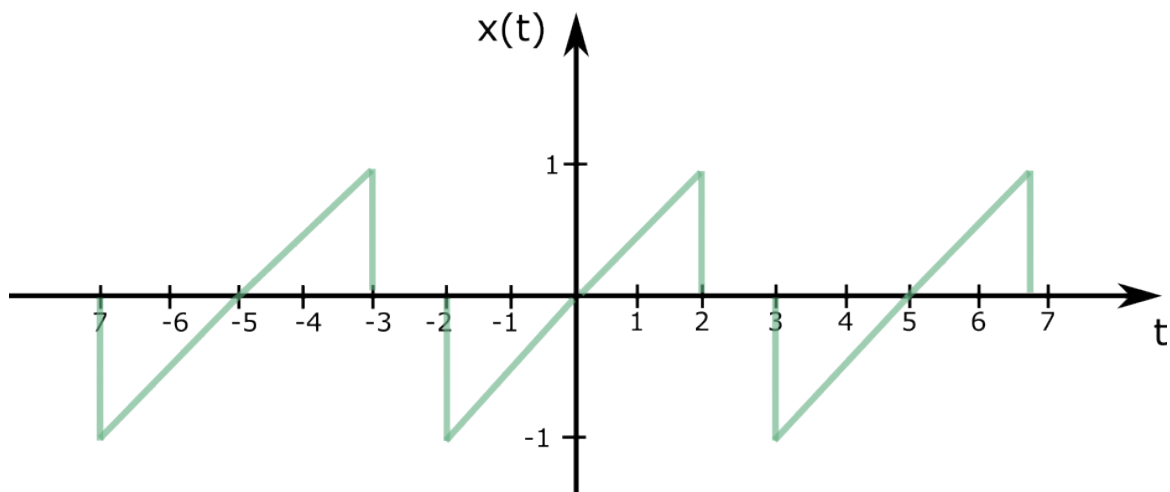
(2) Encontrar el período fundamental y la serie de Fourier asociada para las siguientes señales graficadas considerando que son periódicas:

1.



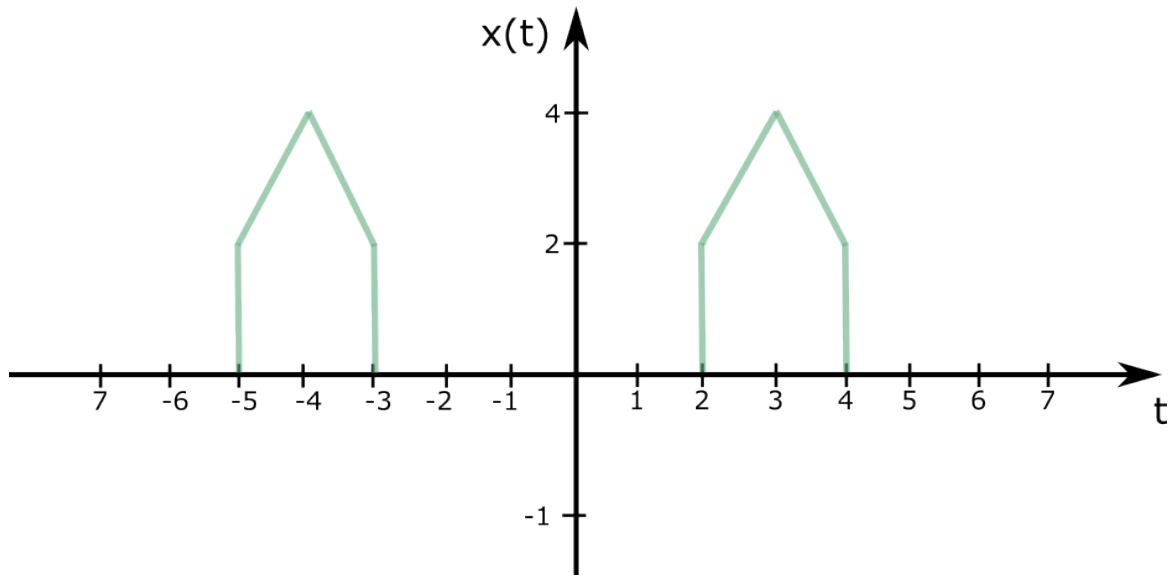
$$a_k = \frac{10 \operatorname{sen}^2(k\pi/5)}{(k\pi)^2}, \quad a_0 = \frac{2}{5}, \quad T = 5, \quad \omega_0 = \frac{2\pi}{5}$$

2.



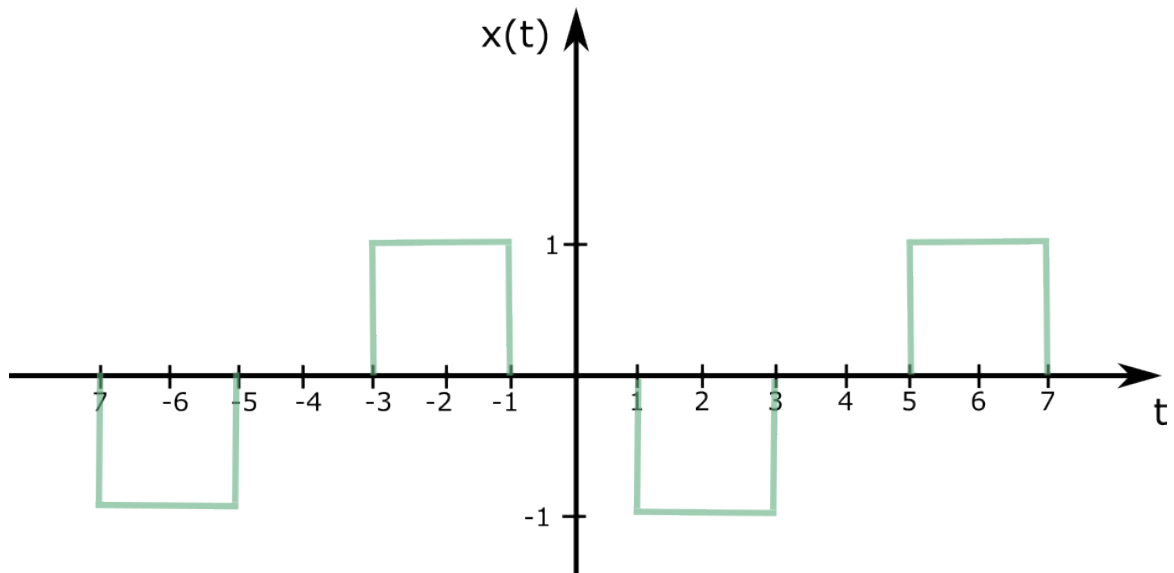
$$a_k = j \frac{1}{5} \left[\frac{3 \cos(k2\pi/3)}{k\pi} - \frac{9 \operatorname{sen}(k2\pi/3)}{(k\pi)^2} \right], \quad a_0 = 0, \quad T = 5, \quad \omega_0 = 2\pi/5$$

3.



$$a_k = e^{-jk6\pi/7} \left[\frac{\text{sen}(k2\pi/7)}{k\pi} + \frac{98\text{sen}^2(k\pi/7)}{(k\pi)^2} \right], \quad a_0 = 8, \quad T = 7, \quad \omega_0 = 2\pi/7$$

4.

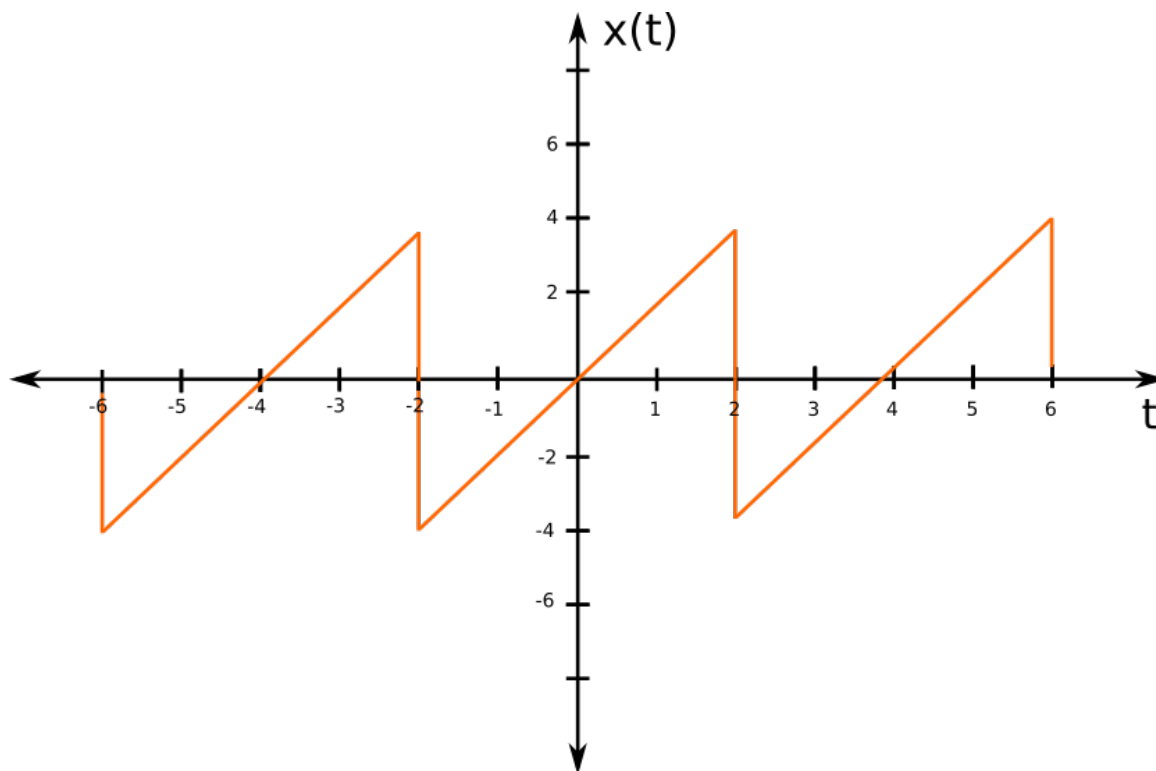


$$a_k = j2 \frac{\text{sen}^2(k\pi/4)}{k\pi}, \quad a_0 = 0, \quad T = 8, \quad \omega_0 = \pi/4$$

Tema 2: Sistemas SLIT y Serie de Fourier

(1) Encontrar los coeficientes a_k , b_k de la serie de Fourier que representa la salida de los sistemas expresados y verifique Parseval

1.

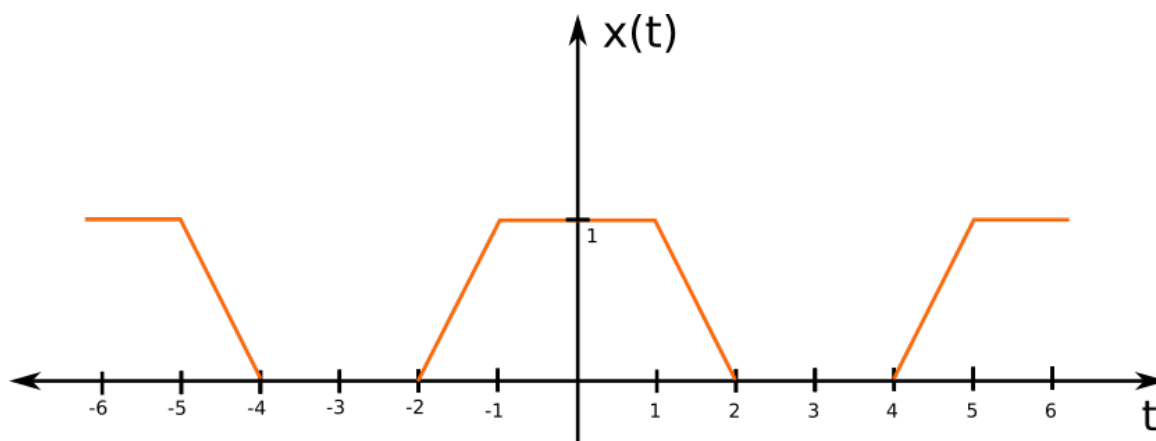


con $h(t) = e^{-4|t|}$

$$a_k = j \frac{2(-1)^k}{k\pi}, \quad a_0 = 0$$

$$b_k = \frac{16}{32 + (k\pi)^2} j \frac{2(-1)^k}{k\pi}$$

2.

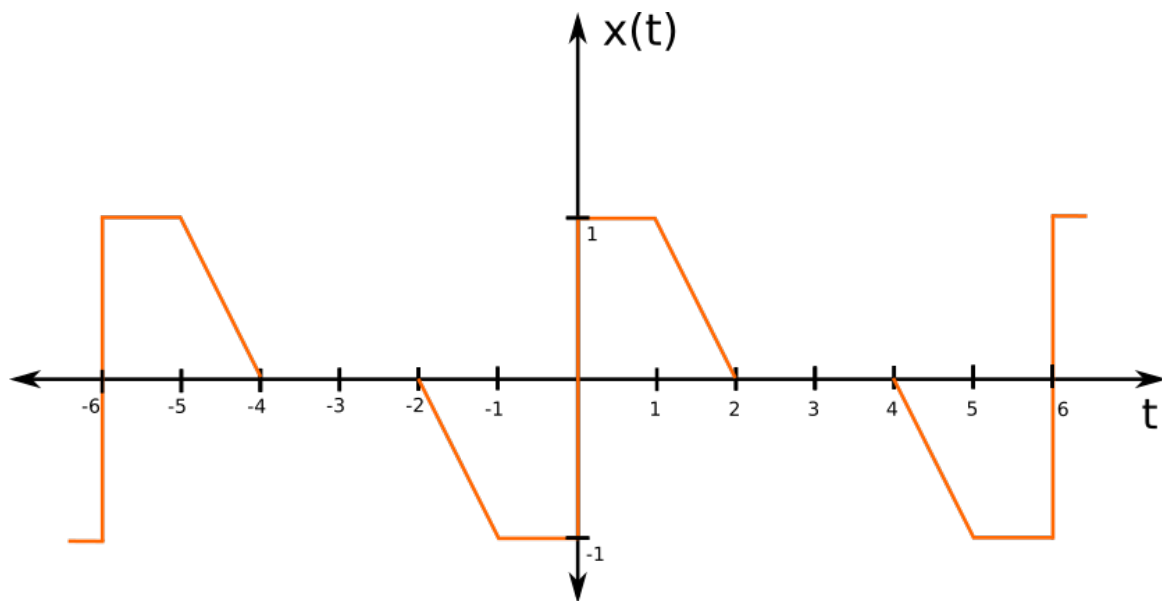


$$\text{con } h(t) = \mu(t) \cdot e^{-2t}$$

$$a_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2)$$

$$b_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) \frac{2}{4 + jk\pi}$$

3.

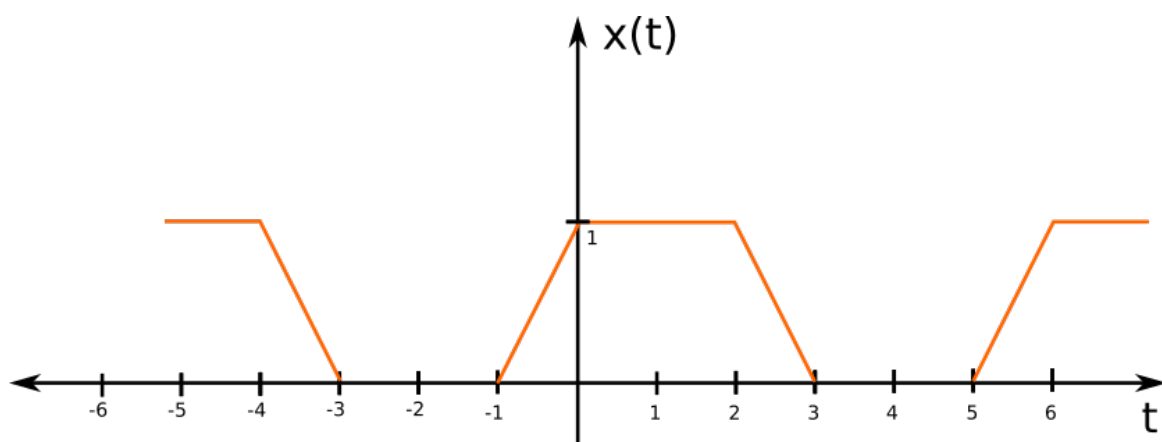


$$\text{con } h(t) = \delta(t - 3) + \delta(t + 3)$$

$$a_k = j \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \cos(k\pi/2)$$

$$b_k = 2(-1)^k a_k$$

4.



$$\text{con } h(t) = \mu(t) - \mu(t-1)$$

$$a_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) e^{-jk\pi/3}$$

$$b_k = a_k \frac{8 \text{sen}(k\pi/4)}{k\pi} e^{-jk\pi/6} = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) e^{-jk\pi/3} \frac{8 \text{sen}(k\pi/4)}{k\pi} e^{-jk\pi/6}$$